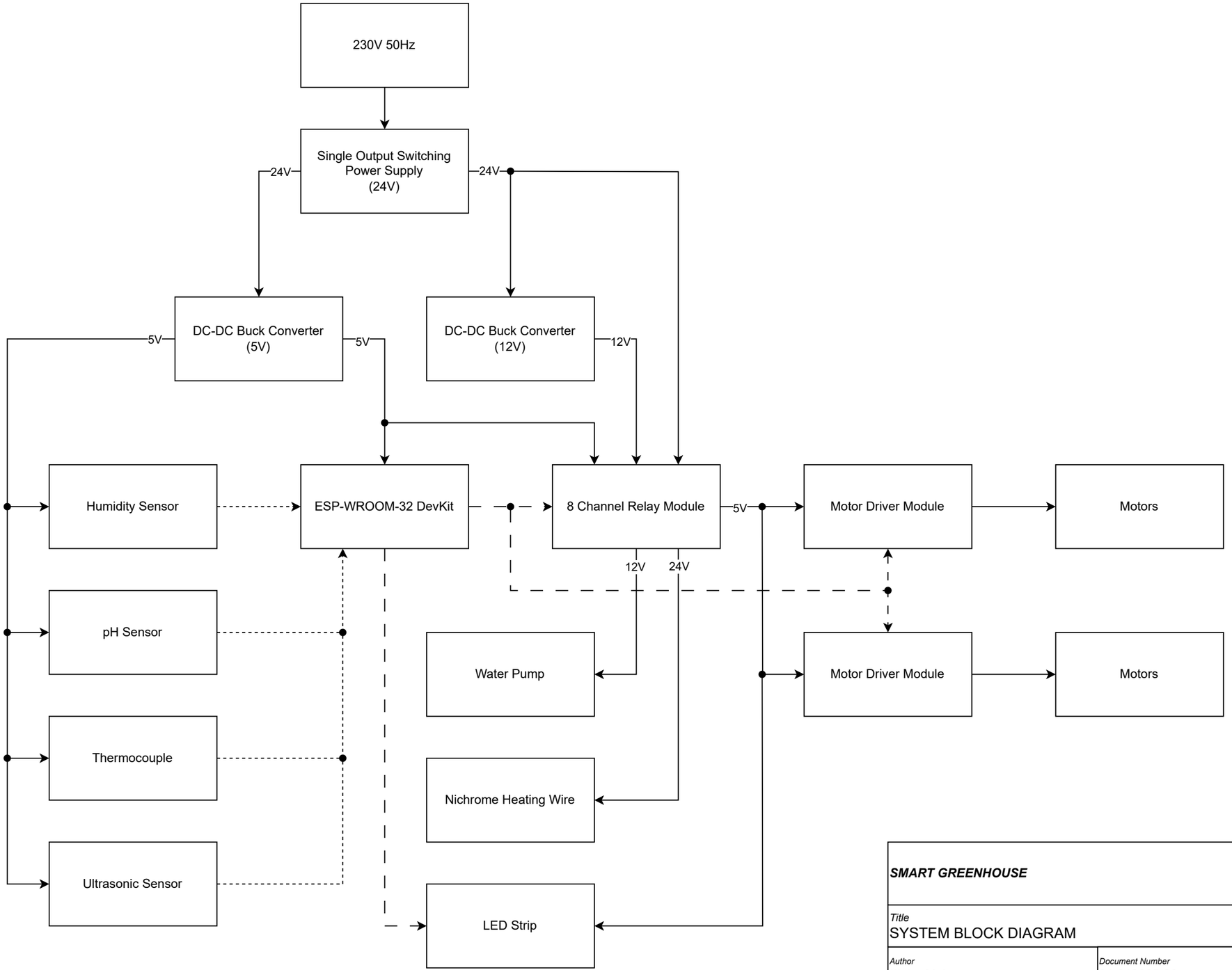
ZOVUYO INNOVATIONS

SMART GREENHOUSE ELECTRICAL AND ELECTRONICS

Author	*Supervisor*	*Facilitator*
Sipho Mokoena	Omuhle Wanda	Oluhle Jawe
Document Number 01-012025-00		*REV* v1.0

Last Modification Date: Tuesday, 01 July 2025



SMART GREENHOUSE

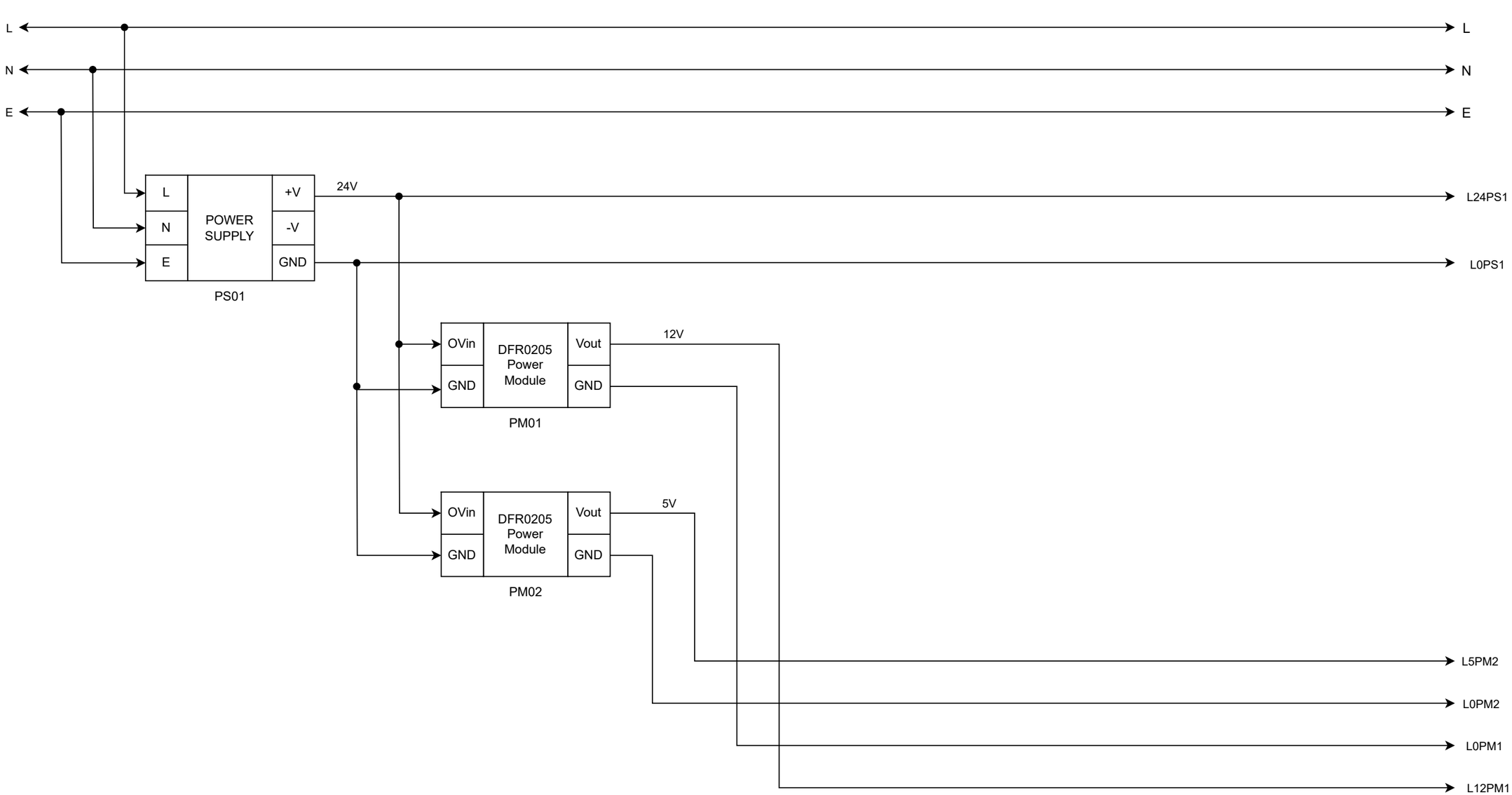
ZOVUYO INNOVATIONS

Title
SYSTEM BLOCK DIAGRAM

Author	*Document Number*	*REV*
Sipho Mokoena	01-012025-00	v1.0

Date: Tuesday, 01 July 2025

SHEET 1 OF 6



SMART GREENHOUSE

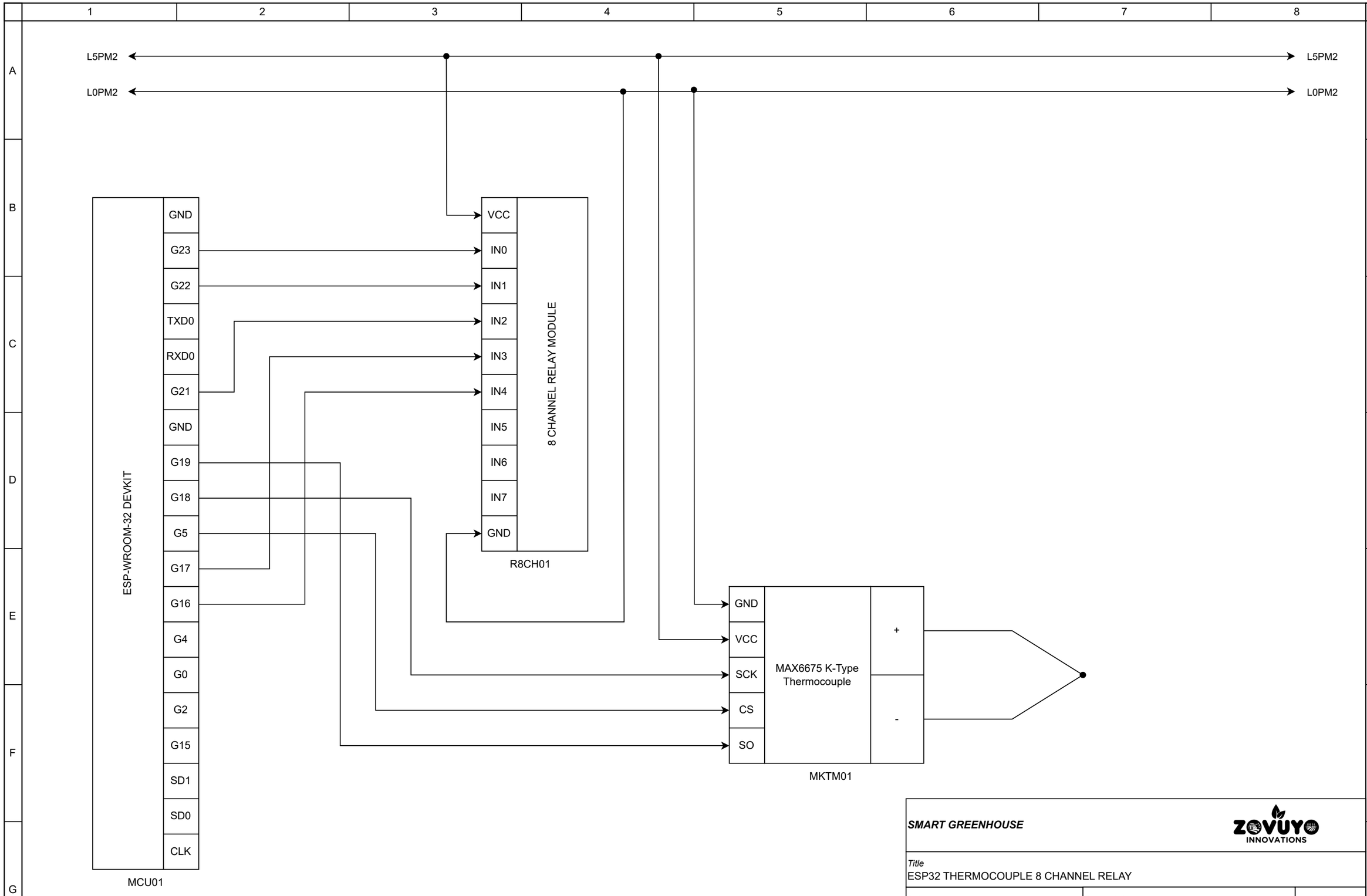
ZOVUYO INNOVATIONS

Title
POWER

Author	*Document Number*	*REV*
Sipho Mokoena	01-012025-00	v1.0

Date: Tuesday, 01 July 2025

SHEET 2 OF 6



SMART GREENHOUSE

ZOVUYO INNOVATIONS

Title
ESP32 THERMOCOUPLE 8 CHANNEL RELAY

Author	*Document Number*	*REV*
Sipho Mokoena	01-012025-00	v1.0

Date: Tuesday, 01 July 2025 — **SHEET 3 OF 6**

L5PM2

L0PM2

L5PM2

L0PM2

ESP-WROOM-32 DEVKIT

3V3
EN
VP
VN
G34
G35
G32
G33
G25
G26
G27
G14
G12
GND
G13
SD2
SD3
CMD
5V

MCU01

VCC
DATA
GND

DHT11
Temperature &
Humidity Sensor

DTHS01

VCC
GND
AOUT

SEN0161
pH Sensor

SPS01

VCC
TRIG
ECHO
OUT
GND

HC-SR04
Ultrasonic
Distance Sensor

HUDS01

1kOhm

2kOhm

SMART GREENHOUSE

ZOVUYO INNOVATIONS

Title
ESP32 HUMIDITY PH ULTRASONIC DISTANCE SENSORS

Author	*Document Number*	*REV*
Sipho Mokoena	01-012025-00	v1.0

Date: Tuesday, 01 July 2025

SHEET 4 OF 6

L12PM1

L0PM1

L12PM1

L0PM1

ESP-WROOM-32 DEVKIT

3V3
EN
VP
VN
G34
G35
G32
G33
G25
G26
G27
G14
G12
GND
G13
SD2
SD3
CMD
5V

MCU01

8 CHANNEL
RELAY MODULE

NO0
COM0
NC0
NO1
COM1
NC1

R8CH01

+12V
GND
+5V
IN1
IN2
IN3
IN4

L298N
Motor Driver

AO1
A02
BO1
B02

LMD01

+12V
GND
+5V
IN1
IN2
IN3
IN4

L298N
Motor Driver

AO1
A02
BO1
B02

LMD02

M
+
-

M
+
-

M
+
-

SMART GREENHOUSE

ZOVUYO INNOVATIONS

Title
ESP32 MOTOR DRIVERS DC MOTORS

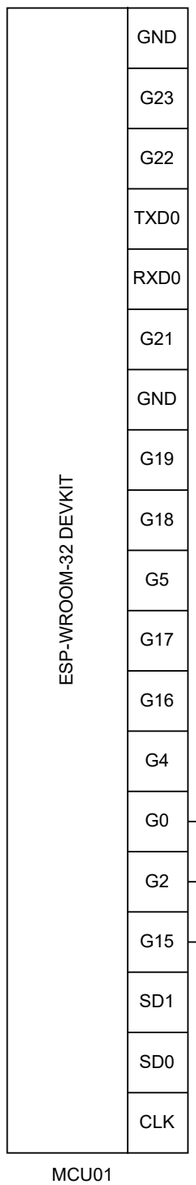
Author	*Document Number*	*REV*
Sipho Mokoena	01-012025-00	v1.0

Date: Tuesday, 01 July 2025

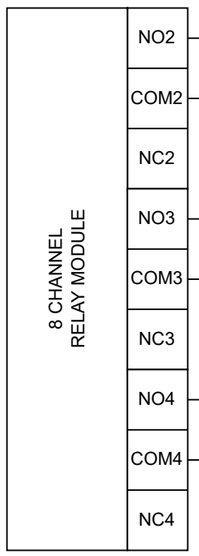
SHEET 5 OF 6

L5PM2
L0PM2
L12PM1
L0PM1
L24PS1
L0PS1

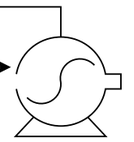
L5PM2
L0PM2
L12PM1
L0PM1
L12PM1
L0PM1



MCU01

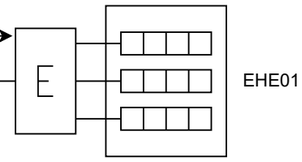


R8CH01



WP01

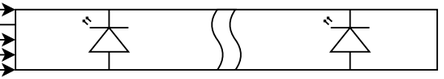
Water Pump



EHE01

Electric Heating Element

LEDS01



LED Strip

SMART GREENHOUSE

ZOVUYO INNOVATIONS

Title
ESP32 8 CHANNEL RELAY LEDS WATER PUMP ELECTRIC HEATING ELEMENT

Author	*Document Number*	*REV*
Sipho Mokoena	01-012025-00	v1.0

Date: Tuesday, 01 July 2025 | **SHEET 6 OF 6**